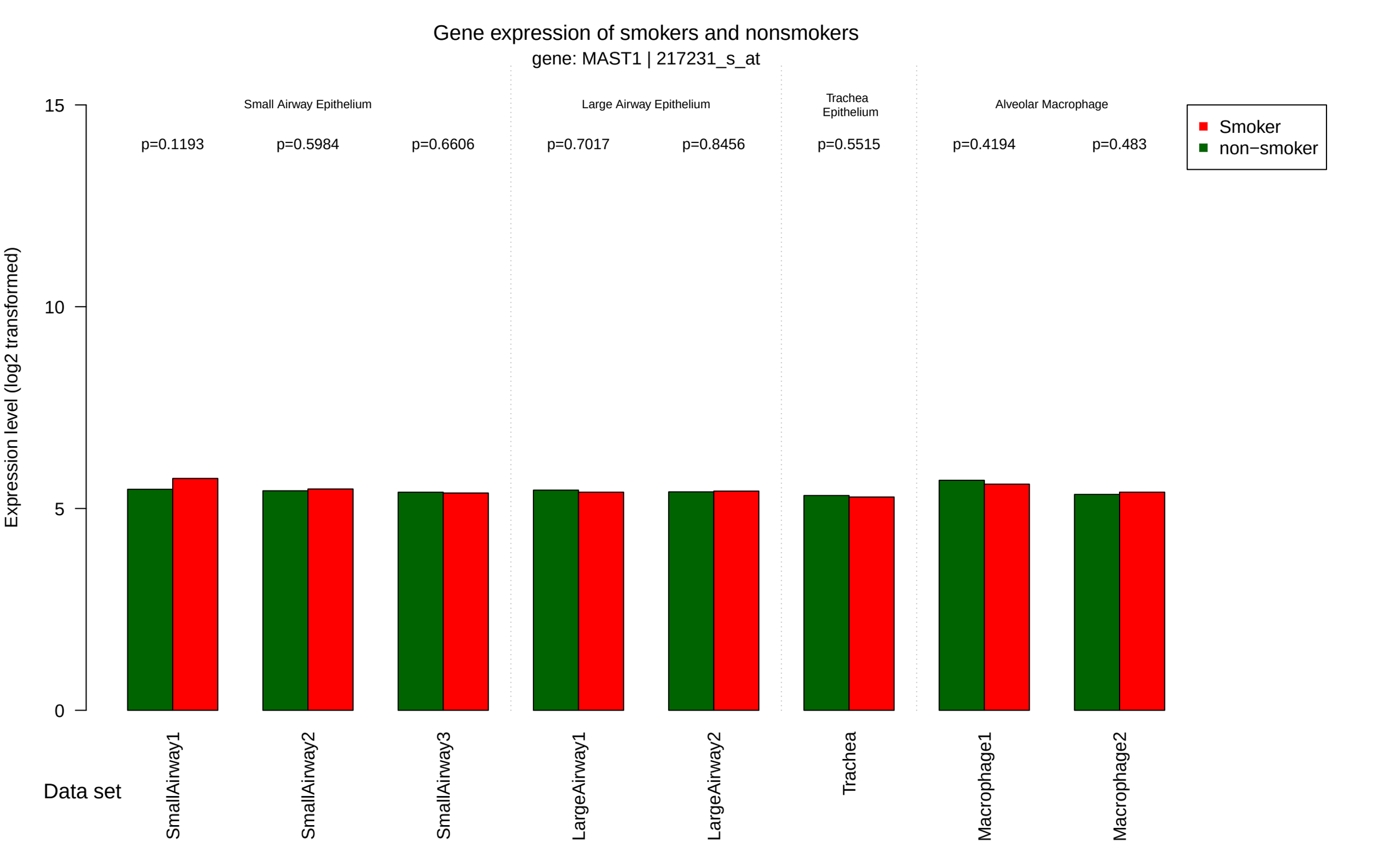
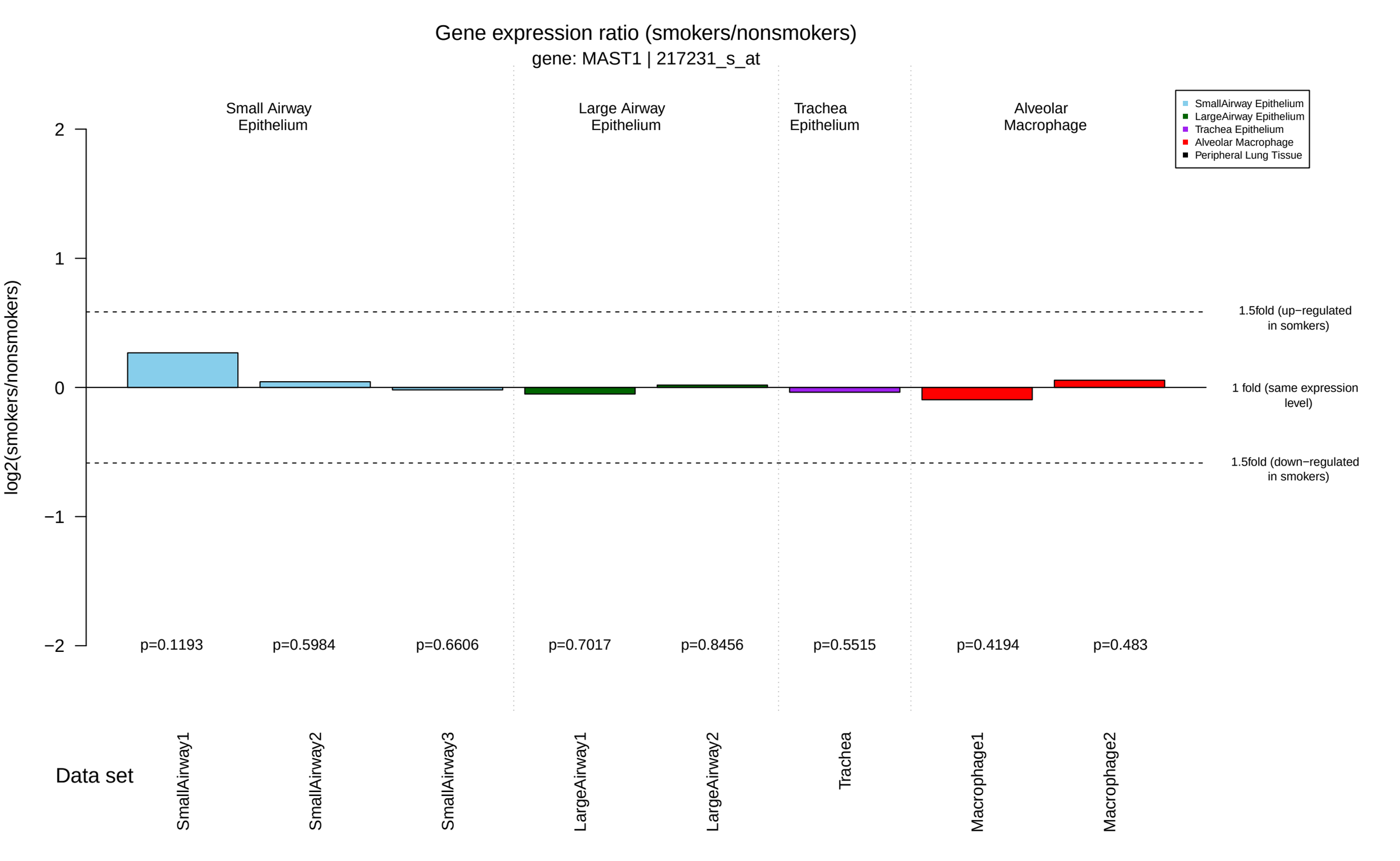
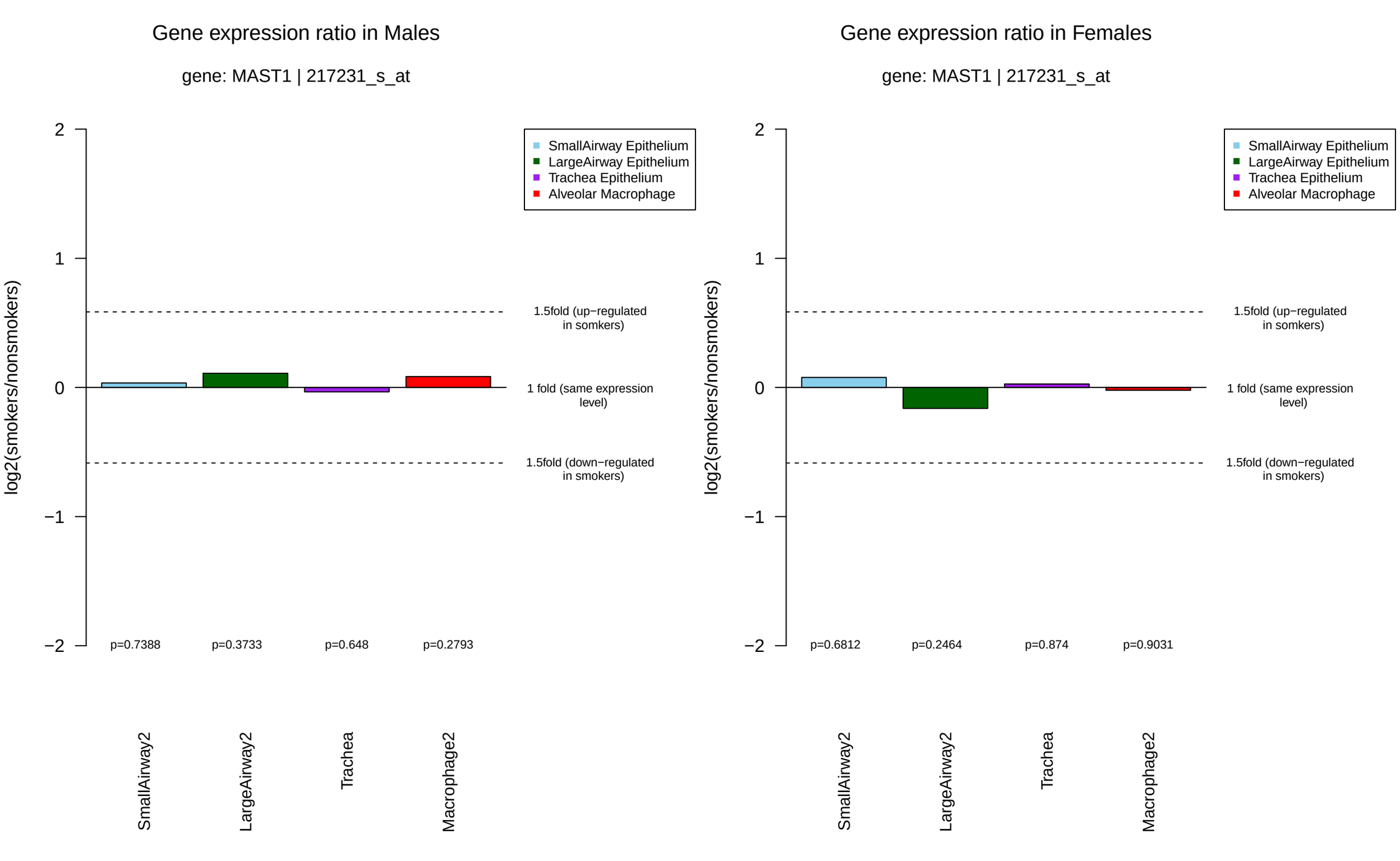
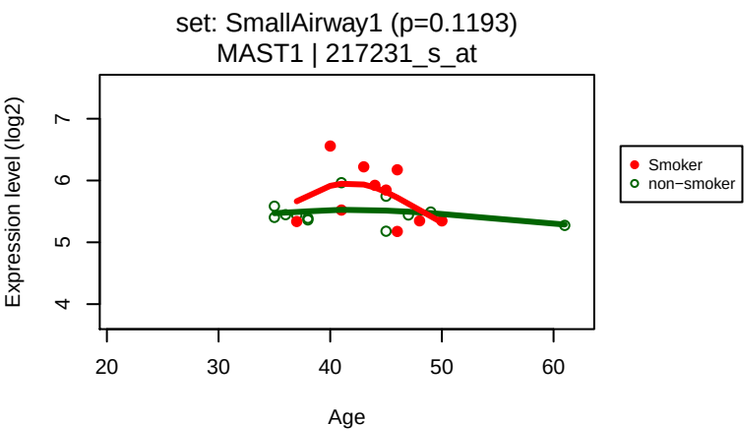
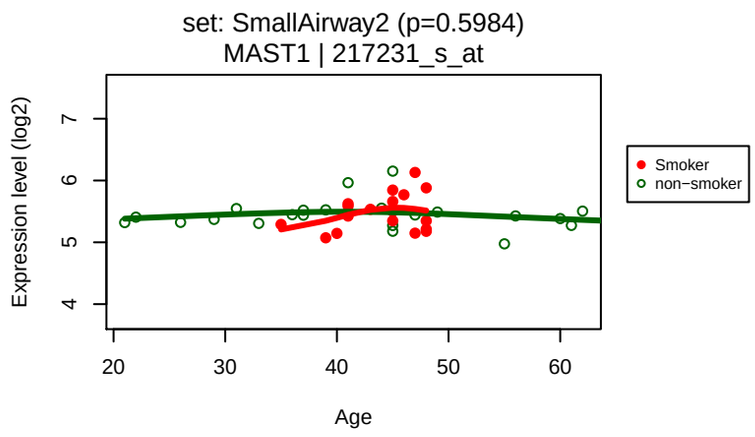
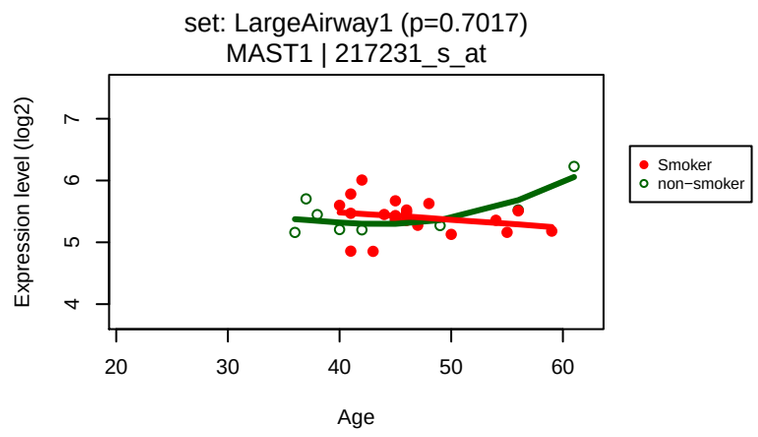

Gene expression of smokers and nonsmokers
gene: MAST1 | 217231_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1193
p=0.5984
p=0.6606
p=0.7017
p=0.8456
p=0.5515
p=0.4194
p=0.483
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

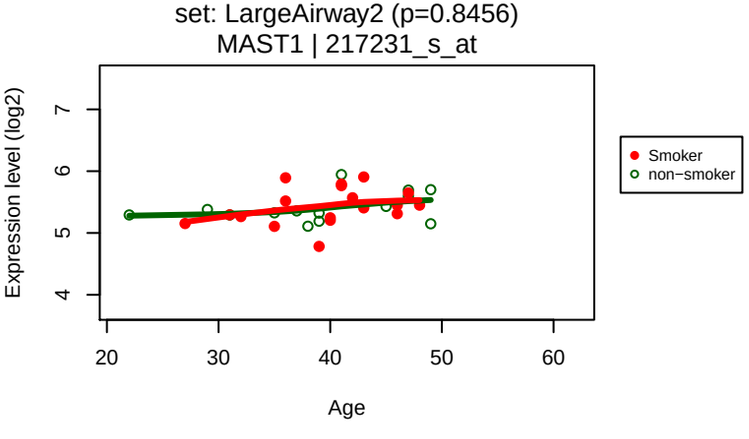

Gene expression ratio (smokers/nonsmokers)
gene: MAST1 | 217231_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1193
p=0.5984
p=0.6606
p=0.7017
p=0.8456
p=0.5515
p=0.4194
p=0.483
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

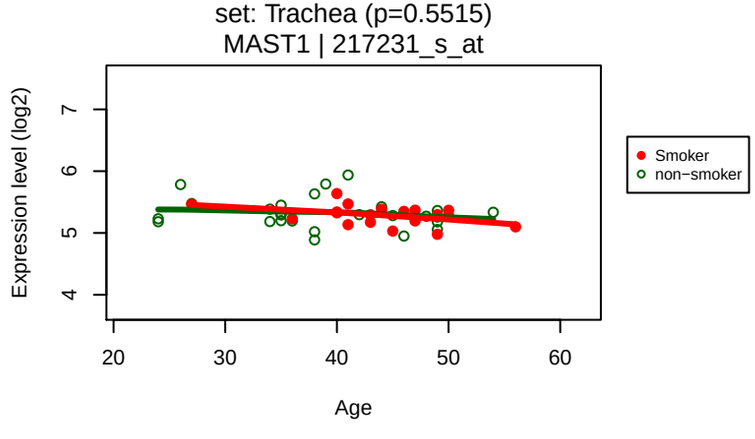

Gene expression ratio in Males
gene: MAST1 | 217231_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7388
p=0.3733
p=0.648
p=0.2793
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MAST1 | 217231_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6812
p=0.2464
p=0.874
p=0.9031
SmallAirway2
LargeAirway2
Trachea
Macrophage2

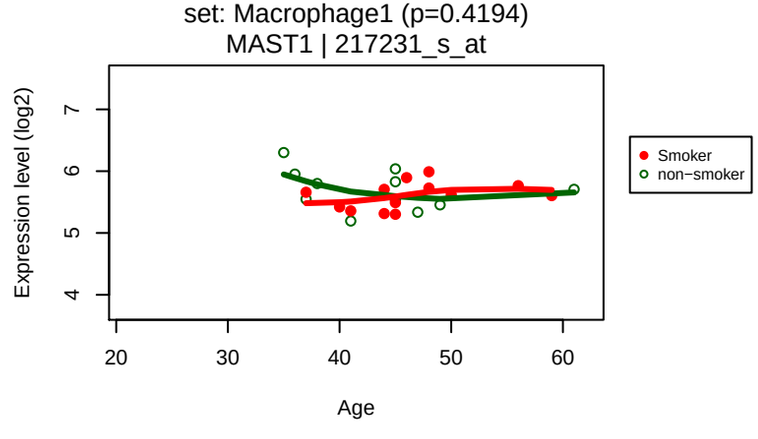

set: SmallAirway1 (p=0.1193)
MAST1 | 217231_s_at
Expression level (log2)
Age
Smoker
non−smoker

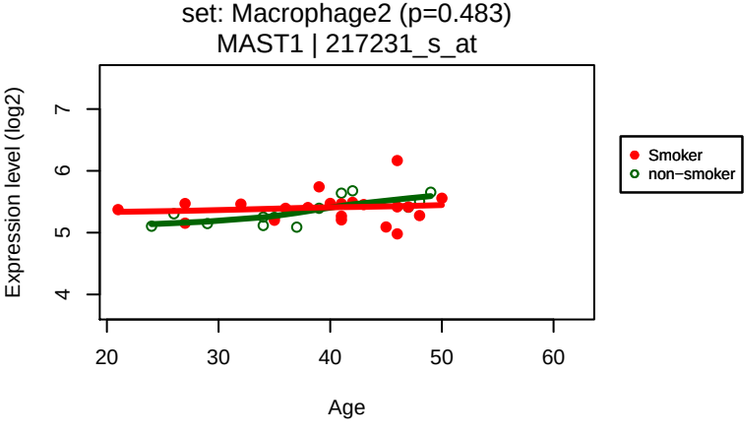

set: SmallAirway2 (p=0.5984)
MAST1 | 217231_s_at
Expression level (log2)
Age
Smoker
non−smoker

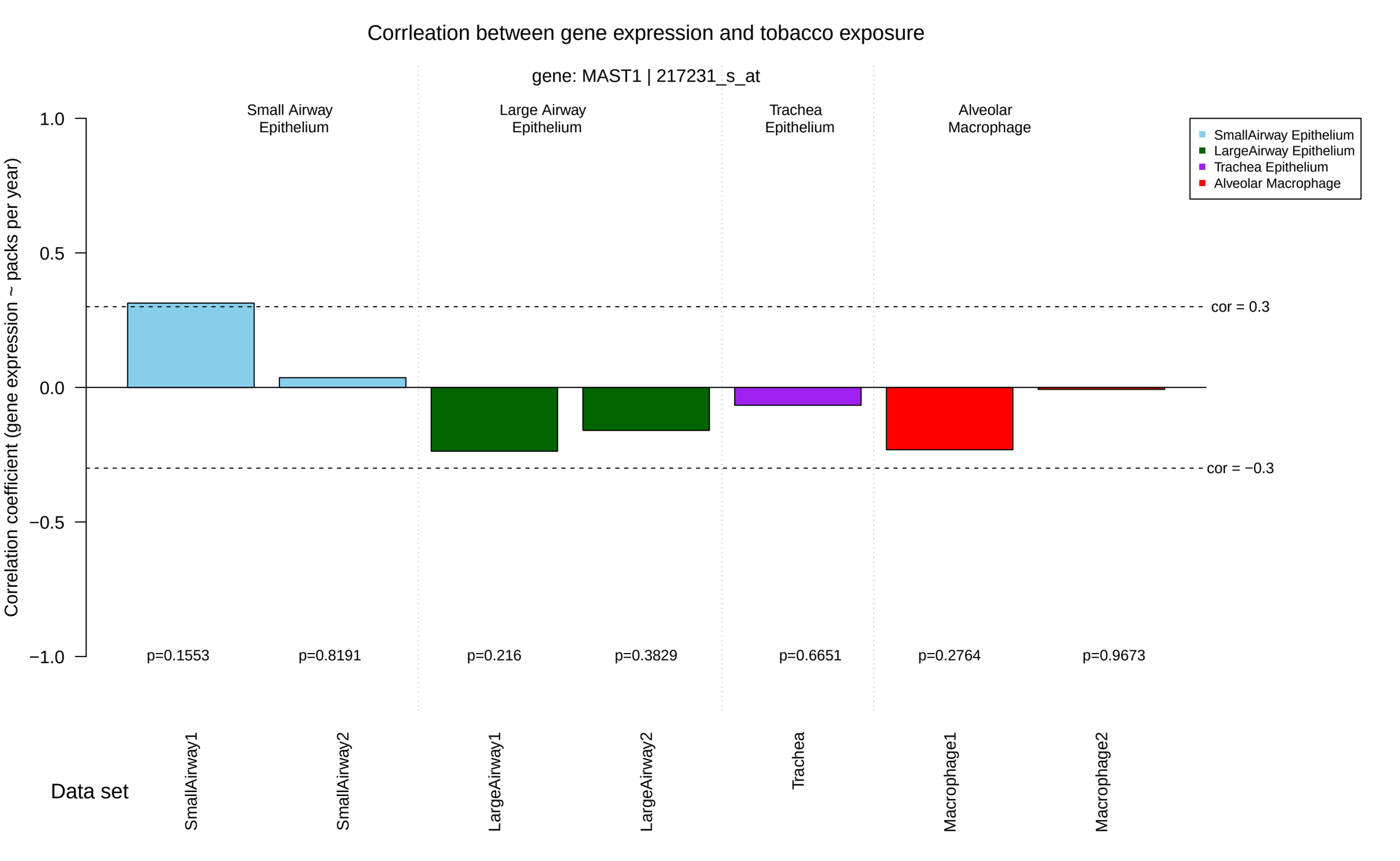

set: LargeAirway1 (p=0.7017)
MAST1 | 217231_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8456)
MAST1 | 217231_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5515)
MAST1 | 217231_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4194)
MAST1 | 217231_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.483)
MAST1 | 217231_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MAST1 | 217231_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1553
p=0.8191
p=0.216
p=0.3829
p=0.6651
p=0.2764
p=0.9673
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2